

# CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

2<sup>nd</sup> Semester of B. Pharm. Examination

University Theory Examination May 2015

PH120 Pharmaceutical Engineering-II

Date: 15-5-2015, Friday

Time: 10:00 a.m. to 1:00p.m.

Maximum Marks: 80

## Instructions:

1. There are three sections in this question paper.
2. SECTION – I comprises of Question 1. Total marks for Section 1 are 20. There are 20 sub-questions (MCQ type). Answers to SECTION – I are to be given in Answer Sheet for MCQ type questions provided to you. Maximum time allotted for SECTION – I is 30 minutes. Answers to SECTION – I must be written during the first 30 minutes of the examination.
3. Answers to SECTION – II and SECTION – III are to be provided in separate Main Answer Books provided to you.
4. Figures to right indicate marks.
5. Draw neat sketches wherever necessary.

## SECTION - I

- Q 1** Attempt all questions. Each question is of one mark. **20**
1. An example of unlimited area of filtration is  
[A] Filter press  
[B] Rotary drum filter  
[C] Both  
[D] None of these
  2. Which of the following is not the filter aid.  
[A] Bentonite  
[B] Kiesulghur  
[C] Talc  
[D] Silica
  3. For what centrifuges cannot be used?  
[A] The separation of well-settling solid particles  
[B] The separation of non-miscible fluids  
[C] The wetting of solid systems  
[D] The separation of solids which settle down with difficulty
  4. A ton of refrigeration (TR) is equivalent to \_\_\_\_\_ Btu/hr of heat.  
[A] 12  
[B] 120  
[C] 1200  
[D] 12,000
  5. HVAC system has to keep positive pressure in  
[A] Commercial kitchens  
[B] Tablet manufacturing area  
[C] Fume hood laboratories

[D] Parenteral department

6. Which instrument is used to measure humidity?  
[A] Thermometer  
[B] Hygrometer  
[C] Rotameter  
[D] Orifice meter
7. In which pharmaceutical process HVAC system required?  
[A] Manufacturing  
[B] Storage  
[C] Packaging  
[D] All
8. Perforated basket centrifuge is \_\_\_\_\_ process instruments.  
[A] Continues  
[B] Batch  
[C] A & B  
[D] None
9. What is the mechanism of surface filtration?  
[A] Straining and impingement  
[B] Entanglement  
[C] Attractive Forces  
[D] All
10. What are the properties of filter aid?  
[A] Non toxic  
[B] Chemically inert  
[C] Low cost  
[D] All
11. The rate of evaporation is  
[A] directly proportional to the temperature of the liquid  
[B] inversely proportional to the temperature of the liquid  
[C] independent of the temperature of the liquid  
[D] directly proportional to the humidity of the surrounding air
12. Distillate is a liquid that  
[A] mixes together  
[B] condenses after distillation  
[C] does not sink due to very fine particles  
[D] a substance that will not dissolve
13. Petroleum products are refined by \_\_\_\_\_ distillation.  
[A] Flash distillation  
[B] Fractional distillation  
[C] Azeotropic distillation  
[D] None of above
14. If a substance decomposes if heated to its boiling point, the suitable method is  
[A] Simple distillation  
[B] Fractional distillation

- [C] Vacuum distillation  
[D] None of above
15. At higher temperture the rate of drying of liquids  
[A] Increases  
[B] Decreases  
[C] Increases and then decreases  
[D] No change
16. The moisture present as the no. of pounds of water per pound of dry solid when the drying limit has been attained by use of air at any given temperature and humidity is called as  
[A] CMC  
[B] EMC  
[C] Total moisture  
[D] Unbound moisture
17. FMC is calculated as  
[A] EMC-Total moisture  
[B] Total moisture- EMC  
[C] Bound moisture- free moisture  
[D] All of the above
18. In the following dryer, Droplets of liquid is sprayed into stream of hot air to obtained solid particles  
[A] Tunnel dryer  
[B] Spray Dryer  
[C] A&B  
[D] None of the above
19. For thermolabile drugs and substance, following one dryer is used  
[A] Vacumme dryer  
[B] Freez dryer  
[C] A&B  
[D] Drum dryer
20. One of the following is not the part of distillation assembly  
[A] Condensor  
[B] Receiver  
[C] Water jacket  
[D] Buckner funnel

## SECTION – II

- Q 2** Attempt any **FOUR** of the following
- |          |                                                                                                |           |
|----------|------------------------------------------------------------------------------------------------|-----------|
| <b>A</b> | Define filtration. Discuss various factors affecting rate of filtration                        | <b>05</b> |
| <b>B</b> | Explain the term centrifugation. Classify them. Draw labeled diagram of centrifugal sedimentor | <b>05</b> |
| <b>C</b> | Write a short note on components of HVAC system                                                | <b>05</b> |
| <b>D</b> | Discuss structure, principle, working & mechanism of Plate and Frame filter press              | <b>05</b> |
| <b>E</b> | Write a short note on mechanism of filtration.                                                 | <b>05</b> |
| <b>F</b> | Write a short note on refrigeration cycle.                                                     | <b>05</b> |

### SECTION – III

**Q 3** Attempt any **FOUR** of the following

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|----------|-----------------------------------------------------------------------------------------------------------------------------------------|-----------|
| <b>A</b> | Explain principle, working, advantages and disadvantages of falling film evaporator with suitable figure                                | <b>05</b> |
| <b>B</b> | Classify evaporators. Draw the figures of Multiple effect evaporators                                                                   | <b>05</b> |
| <b>C</b> | Enlist the factors affecting evaporation. How temperature and surface area influence the evaporation process ?                          | <b>05</b> |
| <b>D</b> | Explain bubble cap plate column with suitable diagram, Write its advantages and disadvantages                                           | <b>05</b> |
| <b>E</b> | Explain Steam distillation with suitable diagram. Write its application                                                                 | <b>05</b> |
| <b>F</b> | Define azeotrope. Give example of Azeotrope with their proportion. Describe the method by which the Azeotropes are getting distillation | <b>05</b> |

**Q 4** Attempt any **FOUR** of the following

- |          |                                                                                                                                               |           |
|----------|-----------------------------------------------------------------------------------------------------------------------------------------------|-----------|
| <b>A</b> | Describe principle and working of Freez dryer                                                                                                 | <b>05</b> |
| <b>B</b> | Explain working advantages and disadvantages of any one dryer which work on the principle of convection<br>Draw the neat diagram of following | <b>05</b> |
| <b>C</b> | A. Fluid bed dryer<br>B. Drum Dryer                                                                                                           | <b>05</b> |
| <b>D</b> | What is fractionation (Rectification) distillation? Explain it with suitable figure                                                           | <b>05</b> |
| <b>E</b> | Explain any one centrifuge which work on principle of batch process.                                                                          | <b>05</b> |
| <b>F</b> | Write a note on Humidity chart with reference to WBT and DBT                                                                                  | <b>05</b> |
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